Synopsis

On

"EYECLINIC: ONLINE EYE CARE AND APPOINTMENT MANAGEMENT SYSTEM"

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1. Introduction

1.1 Online Eye Care and Appointment Management System

EyeClinic is a web-based platform that brings the main services of an eye-care clinic into one application: online appointment booking, doctor consultation workflows, prescription management and an eyewear store. Patients can register, verify their identity through a one-time password (OTP), choose a date and time slot, browse eyewear by category (Men, Women, Kids), keep lens-prescription values with their cart and complete a purchase through a checkout page. Doctors see only the appointments assigned to them and can add prescriptions or mark visits complete, while administrators manage appointments, products, orders, team profiles and patient reviews from a single dashboard.

The system follows a three-tier architecture in which a responsive HTML5, CSS3 and JavaScript front end communicates with an application layer (Node.js/PHP) that applies business rules and exposes API endpoints, and a MySQL database stores users, slots, appointments, products, orders and reviews. Access is controlled through role-based permissions for the Admin, Doctor and Patient roles. A working prototype of the system has already been developed and demonstrated through its Admin, Doctor and Patient interfaces; this synopsis presents the problem, the supporting literature, the objectives and the methodology followed to complete and evaluate the project.

1.2 Problem Statement

Many small and mid-sized eye clinics and optical shops still schedule appointments by phone or walk-in, keep prescriptions on paper and sell eyewear through a separate counter or a separate website. This produces long waiting times, double-booked or missed slots, lost or illegible prescriptions, and no single record that links a patient’s consultation with the glasses they later buy. Existing standalone booking tools and eyewear e-commerce sites each solve only one part of the problem and do not share data with one another.

The problem addressed by this project is therefore to design, implement and evaluate a secure, role-based and easy-to-use web application that integrates appointment scheduling, doctor and prescription workflows and eyewear e-commerce in one platform, while protecting patient data and reducing manual administrative effort.

1.3 Significance of the Study

Eye-care providers need to reduce administrative work and give patients a convenient way to book and follow up. An integrated platform helps in four ways. First, real-time slot availability removes phone-based scheduling and reduces clashes. Second, digital prescriptions linked to a patient account lower the risk of transcription errors and make repeat orders easier. Third, a single admin dashboard gives clinic staff one place to monitor appointments, stock, orders and reviews. Fourth, because the application handles health-related personal data, the study gives practical attention to authentication, OTP verification, password hashing and role-based access control, which are often neglected in small-scale clinic software.

The project also has educational value for the team, since it applies software-engineering practice (requirements analysis, layered design, database design, testing and usability evaluation) to a real-world healthcare-retail problem.

1.4 Novelty of the Study

The novelty of EyeClinic lies in combining, within one codebase and one database, three things that are usually delivered separately: (i) a clinical workflow of slot-based appointments, doctor assignment and prescription entry; (ii) an eyewear e-commerce workflow with cart, lens-prescription fields and multiple payment options (UPI QR code, Razorpay gateway and cash on delivery); and (iii) an administrative layer that moderates reviews and manages team profiles. Slots are organised by shift (Morning, Afternoon, Evening) with live availability counts, and OTP-based verification is applied on top of conventional login. The planned extensions (AI-based frame recommendation, virtual try-on and video consultation) are identified as future work and are not claimed as completed features.

2. Review of Literature

This chapter reviews the conceptual and technical literature that supports the design of EyeClinic: online scheduling and e-health systems, user acceptance and usability, the web architecture and security practices adopted, and the regulatory context in India. The review is organised thematically, and the gaps it reveals motivate the objectives in Chapter 3.

2.1 Surveys and Conceptual Overviews of Online Healthcare Systems

Standard software-engineering texts describe how information systems should be analysed, designed and tested in layers, and stress that requirements for role-based systems must be captured separately for each class of user [2], [3]. The 4+1 view model proposed by Kruchten presents a system through logical, process, development, physical and scenario views and provides a convenient way to document the EyeClinic architecture [13]. For data management, Silberschatz et al. cover relational design, normalisation, transactions and indexing, which underpin the appointment, order and prescription tables of the platform [1].

On the user side, the Technology Acceptance Model identifies perceived usefulness and perceived ease of use as the main determinants of whether people adopt an information system [7]. Nielsen’s usability heuristics and the System Usability Scale give practical ways to design and measure interfaces for non-technical users such as patients and front-desk staff [5], [6]. In India, the Telemedicine Practice Guidelines issued by the Ministry of Health and Family Welfare (2020) set out how remote consultation may be practised, which is relevant to the planned video-consultation extension [18].

2.2 Appointment Scheduling and Clinic–Commerce Systems

Appointment-booking applications and clinic-management software commonly provide slot selection, reminders and status tracking, while eyewear retailers provide catalogues, carts and online payment. In practice these two kinds of application are separate products. A patient who books an eye test on one site and orders glasses on another has two accounts, two records and no link between the prescription and the purchase. The literature on web application design recommends a model–view–controller separation so that scheduling logic, presentation and data access can evolve independently [14], and this separation is used in EyeClinic to keep the appointment and cart modules independent while sharing one user and prescription record.

2.3 Web Technologies, Security and Quality Frameworks

The REST architectural style described by Fielding defines how client–server interactions over HTTP can be organised using stateless requests and uniform resource identifiers, and is the basis of the API endpoints used between the front end and the application layer [4]. Practical references on HTML and CSS, JavaScript, PHP with MySQL and Node.js describe the technologies used in the three tiers [15], [16], [17], [19].

Because the system stores personal and health-related data, security literature is central. The OWASP Top Ten lists the most critical web-application risks, including injection, broken access control and cross-site scripting, and recommends parameterised queries, output encoding and strict authorisation checks [10]. Provos and Mazières describe the adaptive bcrypt password-hashing scheme, which is suited to storing user passwords [11]. RFC 4226 specifies one-time password generation, which informs the OTP verification step [12]. ISO/IEC 25010 defines the quality characteristics (functional suitability, performance efficiency, usability, reliability, security, maintainability and others) against which the finished system can be evaluated [8], and WCAG 2.1 provides accessibility criteria for the user interface [9]. In the Indian context, the Digital Personal Data Protection Act, 2023 sets obligations for processing personal data, which the project takes into account when deciding what data to collect and store [20].

2.4 Research Gaps

The reviewed material shows the following gaps that EyeClinic addresses:

(i)Appointment and clinic-management tools and eyewear e-commerce tools are generally built and sold separately, so prescription, consultation and purchase data are not linked.

(ii)Small clinics lack a lightweight system with distinct, enforced Admin, Doctor and Patient workflows that also applies OTP verification and standard web-security controls.

(iii)Usability and quality evaluation of such integrated systems is rarely reported against recognised instruments (SUS, ISO/IEC 25010); this project plans such an evaluation.

(iv)Advanced features such as AI-based recommendation, virtual try-on and telemedicine are rarely combined with a basic, deployable clinic platform; the project provides the base on which they can be added.

3. Objectives of the Study

The main aim of the study is to develop and evaluate an integrated online eye-care and appointment management system. The specific objectives are:

1.To study existing appointment-scheduling, clinic-management and eyewear e-commerce systems and identify gaps.

2.To design a three-tier architecture and relational database for users, appointments, slots, prescriptions, products, orders and reviews.

3.To implement secure multi-role authentication (Admin, Doctor, Patient) with OTP verification and role-based access control.

4.To develop a slot-based appointment module with real-time availability, doctor assignment and status tracking.

5.To develop an eyewear catalogue with cart, lens-prescription fields and checkout supporting UPI, Razorpay and cash on delivery.

6.To provide an admin dashboard for managing appointments, products, orders, team members and reviews, and a doctor portal for prescriptions.

7.To test the system for functionality, security and usability, and evaluate it against ISO/IEC 25010 characteristics and the System Usability Scale.

8.To document future enhancements such as mobile applications, AI-based recommendation, virtual try-on and video consultation.

4. Methodology

4.1 Research Design

The project follows an applied, design-and-development research approach using an iterative (incremental) software development life cycle. Each iteration covers requirements, design, implementation and testing of one group of modules, and feedback from team members and the guide is used to refine the next iteration [2], [3]. The work is divided into the following phases.

Phase 1 – Requirement analysis. Existing clinic workflows and similar systems are studied, and functional and non-functional requirements are written for the three user roles.

Phase 2 – System design. The three-tier architecture, database schema (ER diagram), use-case, sequence and data-flow diagrams, and the user-interface layouts are prepared.

Phase 3 – Implementation. The front end is built with HTML5, CSS3 and JavaScript; the application layer with Node.js/PHP; and the data layer with MySQL. Modules are developed in the order authentication, appointments, catalogue, cart and checkout, admin panel and doctor dashboard.

Phase 4 – Testing and evaluation. Functional, security and usability tests are carried out as described in Section 4.5.

Phase 5 – Documentation and dissemination. The project report, presentation, exhibition prototype and poster are prepared, and a research paper in IEEE format is written from the results, as required by the department guidelines.

4.2 System Architecture

The system uses a three-tier architecture. The presentation layer delivers responsive pages to the browser. The application layer authenticates users, enforces role permissions, applies business rules (for example, preventing a slot from being booked twice) and exposes REST-style endpoints [4]. The data layer is a MySQL database accessed only through parameterised queries.

4.3 System Modules

Module	Function
Authentication	Registration, login, OTP verification, session management, profile and password management
Appointment	Date and shift-wise slot selection, availability count, booking, approval/rejection, doctor assignment, status tracking
Product Catalogue	Eyewear listing by category (Men/Women/Kids), brand, price, description and image; wishlist
Cart and Checkout	Cart with quantity management, lens-prescription fields, shipping details, UPI/Razorpay/COD payment
Admin Panel	Dashboard metrics and management of appointments, glasses, orders, team members and reviews
Doctor Dashboard	View assigned appointments, add prescriptions, mark appointments complete

Table 4.1: Modules of EyeClinic

4.4 Tools and Technologies

Layer / Purpose	Technology
Presentation	HTML5, CSS3, JavaScript (responsive design)
Application	Node.js
Database	MySQL 8.0+
Web server	Apache 2.4+ or Nginx
Payments	UPI QR code, Razorpay gateway, cash on delivery
Security	Bcrypt password hashing, parameterised queries, input sanitisation and output encoding, CSRF tokens,HTTPS, role-based access control

Table 4.2: Tools and technologies

4.5 Testing and Evaluation Plan

Functional testing will cover each module with test cases for valid and invalid inputs (for example, double booking of a slot, wrong OTP, unauthorised access to an admin page). Security testing will check the controls listed in the OWASP Top Ten that are relevant to the

application, such as SQL injection, cross-site scripting and broken access control [10]. Usability will be evaluated by asking a sample of volunteer users (patients, staff and doctors) to perform defined tasks and complete the System Usability Scale questionnaire [6], together with task-completion time and error counts [5]. Overall quality will be discussed against the ISO/IEC 25010 characteristics [8] and, for the interface, against WCAG 2.1 accessibility criteria [9]. The results will be reported in the final project report and the research paper.

4.6 Expected Outcomes

The expected outcome is a deployable, role-based web application that supports the complete patient journey from booking to purchase, a tested and documented codebase, evaluation results on functionality, security and usability, a research paper in IEEE format, and a project exhibition prototype with a poster.

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